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Digital hair removal by deep learning for skin lesion segmentation

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ABSTRACT

Occlusion due to hair in dermoscopic images affects the diagnostic operation and the accuracy of its analysis of a skin lesion. Also, dermis hair has the following different characteristics: thin; overlapping; faded; of similar contrast or colour to the underlying skin; and obscuring/covering textured lesions. These make digital hair removal (DHR), which involves hair segmentation and hair gap inpainting, a challenging task. Thus, traditional hard-coded threshold-based hair removal methods are not effective, resulting in over-removal which loses important information of the skin lesion, or under-removal which cannot remove the hair effectively. In this paper, we propose a deep learning approach to DHR based on U-Net and a free-form image inpainting architecture. In hair segmentation, a well-labelled dataset is created and used to train U-Net in order to obtain accurate hair masks. In inpainting, a free-form image inpainting architecture (i.e., Gated convolution and SN-PatchGAN) which has been trained on millions of images is used to inpaint any hair gaps. We also propose an evaluation method to analyze the effect of hair removal based on a single dermoscopic image, named intra structural similarity (Intra-SSIM). The process of DHR is repeated until there is no change in the average value of Intra-SSIM. Using the ISIC 2018 dataset, the performance of the proposed method is shown to be better than other state-of-the-art methods.

1. Introduction

Skin cancer is a major public health problem in the world, and melanoma is the most dangerous type of skin cancer [1]. Although the mortality rate of someone with melanoma is high, if detected early the survival rate exceeds 95%. As pigmented lesions occur on the surface of the skin, melanoma is amenable to early detection by expert visual inspection. The most common method of diagnosing a melanoma is via scanning it. One of the most commonly used diagnostic tool is the ABCDE rule: **A**symmetry, **B**orders (irregular with edges and corners), **C**olour (variegated), **D**iameter (greater than 6 mm), and **E**volving over time [2]. Dermoscopy is a noninvasive skin imaging technique that helps diagnose skin lesions, especially in the diagnosis of melanoma [3]. However, in the publicly available datasets containing 2594 dermoscopic images from ISIC Archive (2018 ISBI challenge), there are 1751 images with hair occlusion, accounting for 67.5% of the dataset. Occlusion due to hair causes segmentation algorithms to fail to identify the correct lesion border, and can cause errors in estimating texture measures [4]. Such a occlusion thus remains a key challenge facing computer aided diagnosis of malignant melanoma.

Although many digital hair removal (DHR) algorithms have been developed to address the above-mentioned problem, these algorithms still have two major problems unresolved: over-removal (false positive) and under-removal (false negative) of the occluding hairs. There are two main reasons for these two problems. First, the artefacts of dermoscopy are diverse. These artefacts can be natural, such as hair and veins with a wide range of characteristics, or can be external, such as bubbles and ruler markers [5], as shown in Fig. 1. These artifacts confuse the processes of melanoma border detection and feature extraction [6]. Second, hard-coded threshold-based algorithms are not generalised and cannot cope with these changing characteristics. Generally, these algorithms primarily solve the removal of only dark thick hair on comparatively light skin [7]. Therefore, a solution that can account for the subtle differences in the pixel values and has a strong generalisation ability is needed. Deep learning has been successfully used to solve many difficult computer vision problems and deep learning models are resilient to high variability in skin images. Thus, they can generalise to different hair types and artefacts in images [5].

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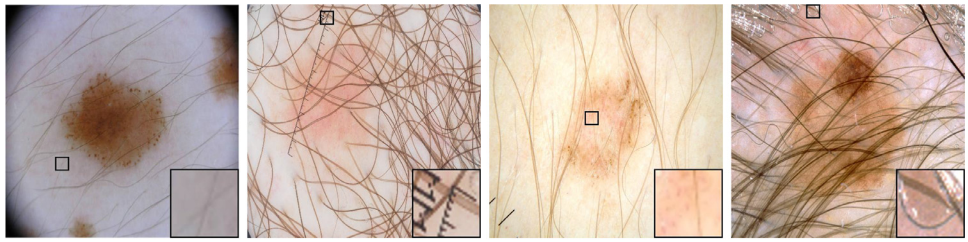

(a) (b) (c) (d)

Fig. 1. Examples of the changing features of dermoscopic images (where the square inset at the bottom-right corner of each image is the zoomed-in enlargement of part of the image denoted by the smaller square), i.e., melanoma with: (a) thin hair, (b) overlapping hair, with ruler marker; (c) of similar contrast and colour to the underlying skin and lesion; and (d) partially occluded textured lesions, with bubbles.

Table 1
Methods for hair segmentation and hair gap inpainting.

Method	Hair detector	Inpainting method	# Test images	Code	Colour Space
DullRazor [8]	Generalized morphological closing	Bilinear interpolation	5	A	Grey
Huang et al. [9]	Multiscale matched filters	Median filtering	20	A	CIE L*a*b
Xie et al. [10]	Adaptive threshold	-	125	A	CIE L*a*b
Xie et al. [11]	Top-hat operator	PED-based [12]	40	N/A	Grey
E-shaver [13]	Prewitt edge	Colour averaging	5	N/A	CIE L*u*v
Schmid-Saugeon et al. [14]	Morphological closing operator	Median filtering	200	N/A	CIE L*u*v
Fiorese et al. [15]	Top-hat operator	PED-based [12]	20	N/A	Grey
Abbas et al. [16]	Derivative of Gaussian	Coherence transport [17]	100	N/A	CIE L*a*b
Koehoorn et al. [18]	Multiscale skeleton Morphological operators	Fast marching [19]	>300	N/A	HSV
Nguyena et al. [4]	Universal matched filter	-	-	N/A	RGB
Zhou et al. [20]	Line detection Curve fitting	Exemplar-based method	460	N/A	CIE L*u*v
Maglogiannis et al. [21]	Bottom-Hat transform, Laplacian of Gaussian, Sobel	Interpolation	10	N/A	Grey
Du et al. [7]	Top-Hat transform Multi-scale curvilinear Matched filtering	-	60	N/A	YUV
P. Bibiloni et al. [22,23]	Soft colour top-hat transforms	Soft colour morphology	2	N/A	CIE L*a*b
Mohamed et al. [5]	Hybrid network	-	>300	N/A	RGB

However, the lack of well-labelled dataset hinders the development of deep learning hair removal techniques.

The contributions of this paper are as follows. We solved: (1) the problem of the lack of hair mask dataset; (2) the problem of poor hair removal; and (3) the difficulty in evaluating the hair removal effect of a single image. Although hair annotation is a time-consuming and tedious task, we have carefully prepared a well-labelled hair mask dataset, named Digital Hair Dataset, as shown in Fig. 4. This dataset is used to train the encoder-decoder segmentation network. The segmentation hair masks produced by the U-Net are then fed to a free-form image inpainting architecture to inpaint the hair gaps. We introduced intra structural similarity (Intra-SSIM) which is designed to evaluate the effect of hair removal of a single image, and use it as a stoppage criteria for DHR. Finally, the input hair image and the inpainted image are presented as inputs to five segmentation algorithms, to demonstrate the importance of accurate hair removal. To our knowledge, this is the first time that a well-labelled hair mask dataset is used to train a convolutional network, and the first time that deep learning is used for hair gap inpainting.

The remainder of the paper is organised as follows. Section 2 reviews the state-of-the-art methods for DHR. Section 3 presents the proposed approach to DHR in detail. Section 4 presents the conducted experiments and the evaluation metrics. Section 5 presents the experimental results and comparison. Finally, Section 6 draws a conclusion from the experimental results.

2. Related work

In the past decade, many methods for hair segmentation and for hair gap inpainting have been developed for DHR. The characteristics of these methods are summarised in Table 1.

2.1. Hair segmentation methods

There are several hair segmentation methods such as mathematical morphology, adaptive thresholding and threshold set representation, edge detection-based, matched filtering, and others [7]. DullRazor [8], the earliest and arguably the best known method, performs morphological operations and utilises three structuring elements at different orientations to detect hair. This method can find dark thick hairs on light skin, but cannot detect low-contrast thin and highly curled hairs. Based on the DullRazor, Schmid-Saugeon et al. [14] convert the image from RGB colour space to CIE L*u*v colour space, and then apply morphological closing operations with a disk structuring element to detect hair. Xie et al. [11] and Fiorese et al. [15] used morphological top-hat operator and hard thresholding to enhance the image and obtain hair masks from the enhanced image. P. Bibiloni et al. [22] used soft colour top-hat transforms method based on black top-hat to extract hair. The black top-hat transform preserves the objects that are smaller than that of the structuring element and those with high contrast with respect to the background [23]. Therefore the top-hat operator is only able to detect dark hair on a light skin.

Koehoorn et al. [18] convert the input image to a luminance threshold-set representation, and apply a general-purpose gap-detection algorithm based on morphological openings/closings and skeletonisation to find thin structures in each thresholded layer. Any potential hairs found in all layers are merged to create a hair mask. Xie et al. [10] enhance an image by simple linear iterative clustering (SLIC) combined with isotropic nonlinear filtering (INF), and then extract hairs by automatic thresholding. Due to the limitation of setting the appropriate threshold, this method cannot fully eliminate dense hairs of varying colour, overlaid on a high-contrast skin texture.

Kiani et al. [13] detect the dominant orientation of hairs in the image using Radon transform, followed by filtering the image with

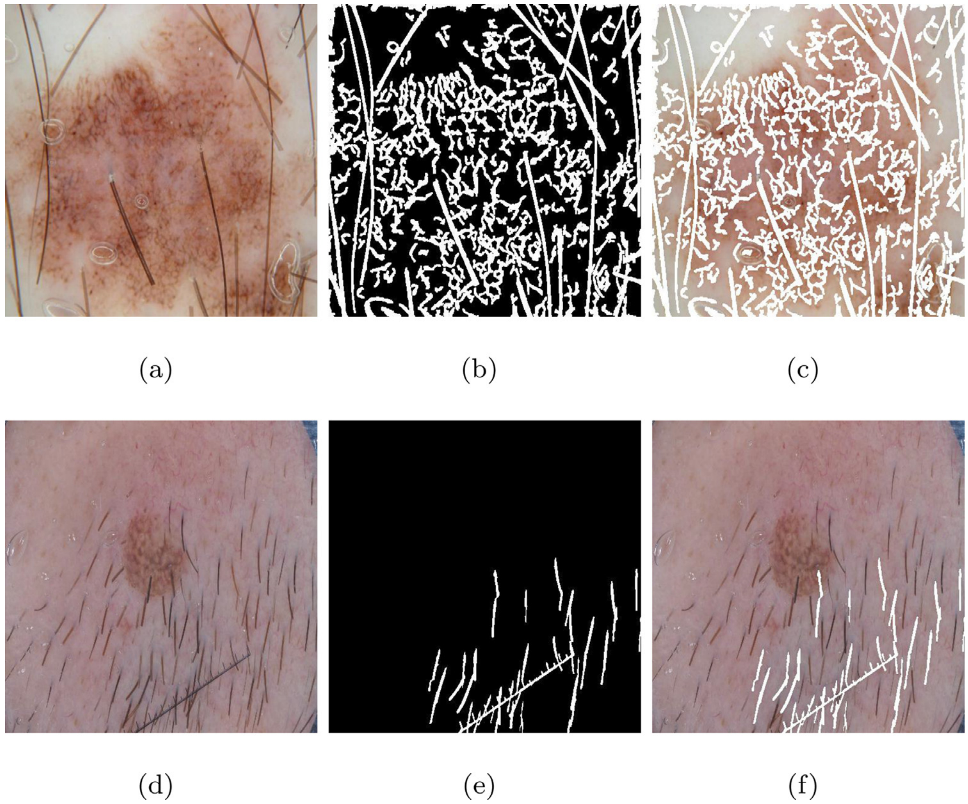

Fig. 2. (a) and (d) are input images. Images (b) and (c) are the hair masks produced by Huang et al. [9] that have many false positives since they treat other textures as hairs. Images (e) and (f) are the hair masks produced by Xie et al. [10] that suffer from false negatives as not all of the hairs are detected.

Prewitt edge detector, called E-shaver, using the orientation of existing hairs. Since the found edges are thresholded, any hair close to skin colour is hard to find [15]. Tossi et al. [24] segment light and dark hairs and ruler markers using adaptive Canny edge detector and refinement via morphological operators. Maglogiannis et al. [21] use an ensemble of edge detectors: Bottom-Hat, Laplacian, Laplacian of Gaussian and Sobel methods. However, the methods are prone to over-segmentation because they attribute all high contrast edges on the skin image to edges of hairs.

Abbas et al. [16] find dark or light hairs using a matched filter with a 1-dimensional Gaussian profile and adaptive threshold. However, a large number of parameters need to be tuned for each input image. Nguyena et al. [4] use a matched filter together with morphological operation and Gaussian curve fitting to detect both dark and light hair without bias. Huang et al. [9] use matched filters to enhance curvilinear structures, and hysteresis thresholding to obtain a hair mask. The method is able to detect small gaps in the hair structure. However, the large number of convolution operations involved increase the computational cost and require long execution time [7].

Zhou et al. [20] detect hair by Steger's [25] line detection algorithm and curve fitting. A hair is detected as points that are linked to obtain long segments. A curve-fit model of the hair is used to discard misclassified pixels. Mohamed et al. [5] were the first to use deep learning to detect hair. They apply a hybrid network of convolutional and recurrent layers for hair segmentation. However, due to the lack of a hair mask dataset, they used weakly labelled data, which weaken the robustness of the network.

Choosing the best threshold and the best colour space is crucial for the accurate segmentation of the above mentioned methods. Therefore, incorrect choice can lead to high rate of false positive and false negative that can change the texture integrity of the lesions, and leave hair traces and artifacts that affect the subsequent diagnosis [5], as illustrated in Fig. 2. Hence, these methods cannot be applied to most images. These only achieve good results on high contrast hair-skin images with thick and dark hairs [7]. Thus, accurate detection of all types of hair remains a challenge.

2.2. Hair gap inpainting

Hair gap inpainting is the second step of DHR. The main methods that have been used for this step include interpolation, median filtering, exemplar-based technique, PDE-based method, colour averaging, coherence transport, and fast marching method [5,6].

The method in [7] replaces the hair pixels by bilinear interpolation and then smooth the final result by an adaptive median filter. A similar interpolation method is used by Nguyena et al. [4]. Maglogiannis et al. [21] replace hair pixels by interpolating in each channel of the RGB colour space. Huang et al. [9] and Schmid-Saugeon et al [14] also apply median filter for hair gap inpainting. However, the use of median filtering eradicates essential subtle elements of the pristine images, making the lesion hazy [6,16]. Exemplar-based method is used for inpainting in [20] to restore the hair gaps with a batch sorting based on some confidence measures. The results show that the method is not only efficient in sample-based texture synthesis and achieves high performance, it also considers the constraints of the surrounding linear structure on the image. However, this method needs to adjust several parameters that are difficult to determine in general [16]. Xie et al. [11] and Fiorese et al. [15] both apply partial differential equation based (PDE-based) methods for inpainting. However, this is an iterative method which cannot guarantee convergence of the parameters involved [5]. E-shaver [13] uses colour averaging which is faster than bilinear interpolation but the repair traces are obvious and unnatural on the inpainted image. Koehoorn et al. [18] remove hairs by using the fast marching method [19], but the use of linear

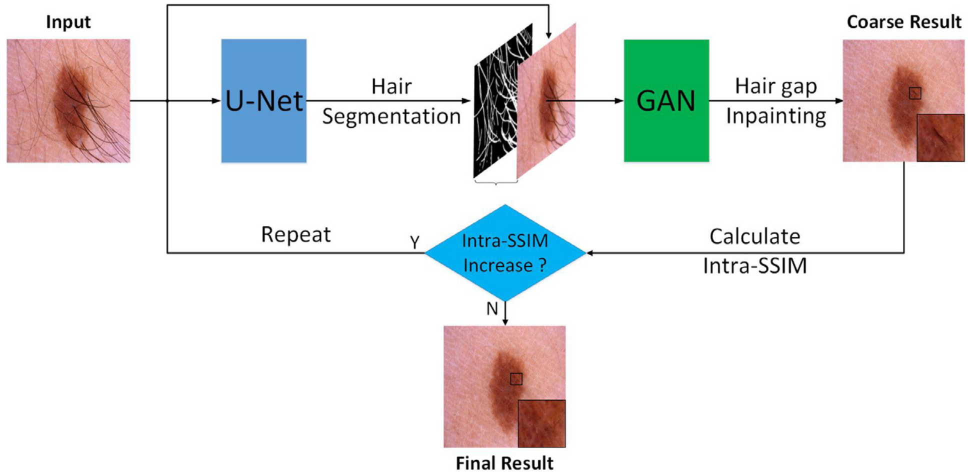

Fig. 3. Overview of the proposed framework with U-Net and GAN for DHR.

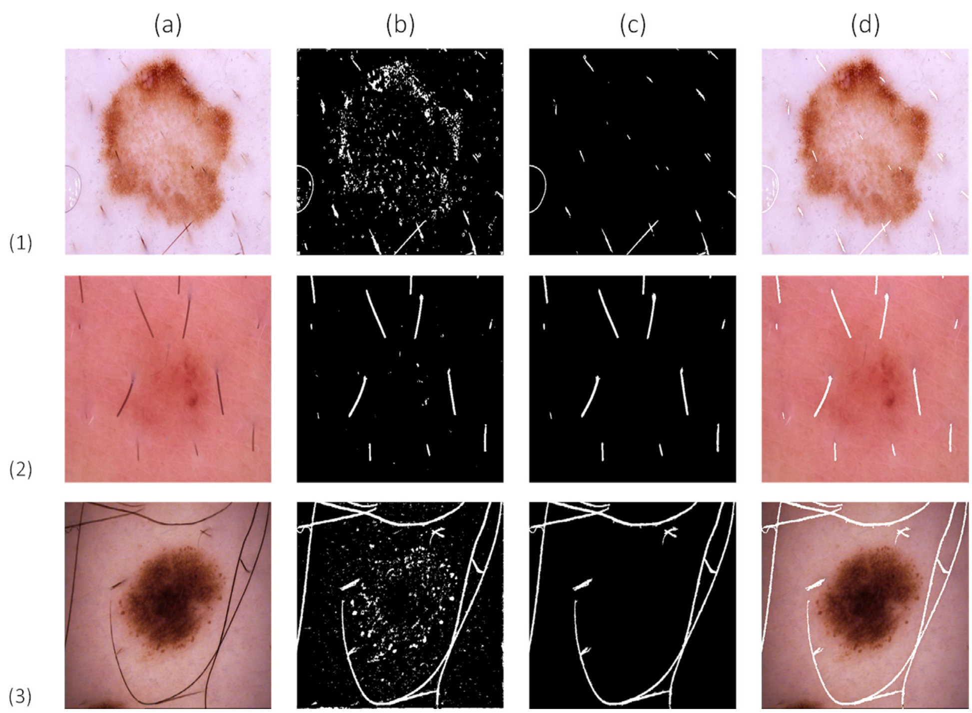

Fig. 4. Examples of creating Digital Hair Dataset: Column (a) are the images from 2018 ISIC dataset. Column (b) the outputs of black top-hat. Column (c) the ground truth created by us by removing over segmented regions. Column (d) the ground truth overlaid on the original images.

and local characteristics results in blurriness when inpainting thick regions. All the above inpainting methods often present undesirable blur and distort the texture of the tumour. Moreover, these methods involve high computational cost.

3. Proposed methods

Deep learning has already made a huge impact in cancer diagnosis due to its distributed data representation with multiple levels of abstraction [26,27]. In this paper, we present a digital hair dataset and apply deep learning to remove hairs in a skin image. Specifically, we propose three innovative methods for DHR, that are used to segment hair, inpaint hair gaps, and evaluate the effect of hair removal, as shown in Fig. 3. First, semantic segmentation framework based on U-Net [28] is used to segment hair in various situations. Second, hair gaps are inpainted by using transfer learning (TL) based on Gated convolution and SN-PatchGAN [29]. Third, our proposed Intra-SSIM is used as a stoppage criteria of the inpainting process and also as a metric to evaluate the effect of hair gap inpainting.

3.1. Creation of digital hair dataset

To train the U-Net model and thus obtain accurate hair masks, we created a digital hair dataset. The process involved: (a) obtaining hair masks by presenting 306 lesion images with hairs from the ISIC 2018 dataset to the existing black top-hat segmentation techniques [23]; (b) careful selection of over segmented hair masks; (c) manual removal of the over segmented regions by a team of four members to create the new Digital Hair Dataset, and finally (d) overlaying the hair masks on the original image to ensure the correctness of the created hair masks. The dataset creators followed the above steps meticulously and the final step (overlaying of annotated data over the original image) in the creation

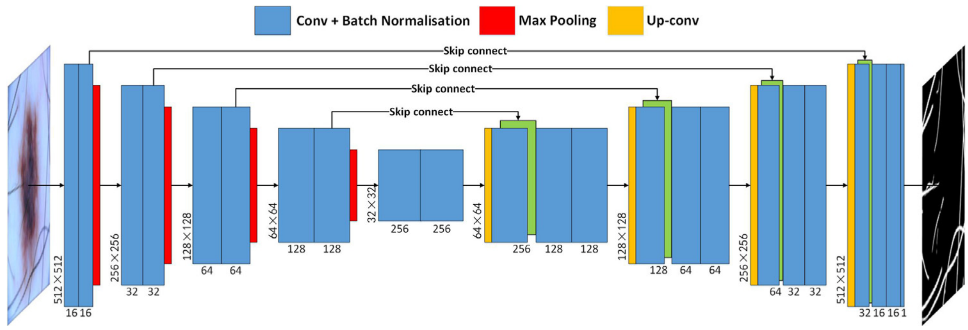

Fig. 5. The U-Net architecture. Each blue box corresponds to a multi-channel feature map obtained by convolving with the corresponding filter kernels. The number of channels involved and the x-y-size of the map are denoted at the bottom and at the lower left edge of the box, respectively. The red box and yellow box respectively correspond to max pooling and transposed convolution, while the green boxes represent feature maps copied through skip connections. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

process was doubled checked to ensure high accuracy and correctness of the hair mask thereby resolving any discrepancies between the hair masks generated by the different creators. The process is illustrated in Fig. 4. Here under-segmented images are avoided since manual recreation of hair marking for under-segmented images is more time consuming and does not ensure accuracy. The hair masks are loaded to kaggle to allow for replicability and for use by the research community.

3.2. U-Net architecture for hair segmentation

The goal of semantic image segmentation is to label each pixel of an image with the corresponding class of what is being represented. Within our context, semantic segmentation is employed for accurate classification of pixels in a skin image into two categories, namely hair and skin. It can better explain the global context of an image [5,30]. Despite the popularity of various traditional computer vision techniques, deep learning architectures, usually Convolutional Neural Networks (CNNs), are surpassing them by a large margin in terms of accuracy and sometimes even efficiency [31]. Thus, many deep learning approaches for medical image segmentation have been introduced [30].

U-Net is one of the most important semantic segmentation frameworks for a CNN which can automatically learn image features. It is widely used in medical image analysis for lesion segmentation, anatomical segmentation, and classification. The advantage of this network framework is that it can not only accurately segment the desired target feature, and effectively process and objectively evaluate medical images, it can also help to improve accuracy in the medical diagnosis [32]. Furthermore, in the case of the lack of training images, U-Net architecture can yield precise segmentation results [28]. Therefore, this architecture is suitable for hair segmentation because of the availability of only small amount of hair dataset.

The U-Net architecture contains two paths as shown in Fig. 5. The first path (left side of the figure) is the contraction path (also called the encoder) which is used to capture the context in the image. It consists of the repeated application of two 3x3 convolutions, each followed by a rectified linear unit (ReLU) and a 2x2 max pooling operation with stride 2 for downsampling, starting from 512x512x3 to 32x32x256. The second path (right side) is the symmetric expanding path (also called the decoder) which is used to enable precise localization via transposed convolutions, starting from 32x32x256 to 512x512x1. Every step in the expanding path consists of an upsampling of the feature map followed by a 2x2 transposed convolution. The transposed convolution halves the number of feature channels via a concatenation with the correspondingly feature map from the contracting path, and two 3x3 convolutions, each followed by a ReLU. At the final layer a 1x1 convolution followed by a sigmoid is used to map each 16-component feature vector to the desired number of classes. The network has 23 convolutional layers, and is an end-to-end fully convolutional network (FCN) [28].

Convolution and pooling down sample an image. Max pooling helps to understand WHAT is there in the image by increasing the receptive field. However, in semantic segmentation it is not just important to know WHAT is present in the image but it is equally important to know WHERE it is present. Hence transposed convolution, which basically learns parameters through back propagation, is chosen to perform up sampling. To get better precise locations at every step of the decoder, skip connections are used to concatenate the output of the transposed convolution layers with the feature maps from the encoder at the same level. After every concatenation two consecutive regular convolutions are applied again so that the model can learn to assemble a more precise output [28,33].

The training model employs Adam as the optimizer and Dice coefficient loss as the loss function. The learning rate is adjusted to 0.1 times of the previous value when the val-loss (the loss of Dice coefficient) is not reduced after 5 consecutive iterations, with a lower limit set to 0.0001. Therefore, by optimizing the learning rate, the model is generalized quickly and accurately. Furthermore, an early stopping mechanism, which terminates the training after 50 consecutive iterations is utilized if the Dice coefficient loss is not reduced, i.e., inferring that the model has been trained to the optimal level. The results of hair segmentation are presented in Section 5.1.

3.3. Transfer learning for hair gap inpainting

The hair segmentation is followed by hair gap inpainting. Due to the lack of skin images, we use Gated convolution and SN-PatchGAN based TL technique to inpaint the hair gap.

Deep learning has achieved significant success in image inpainting [29]. However, due to the lack of a relevant dataset, it has not been used to inpaint hair gap. We are the first to do so. Overcoming the challenge of hair gap inpainting requires a considerable amount of hair free images. However, the current amount of available data is far from sufficient to train a model which will provide good results. We apply TL technique to address the lack of data. TL focuses on storing knowledge gained while solving one problem and applying it to a different but related problem. The purpose of TL is to provide a faster and better solution with less effort to recollect the required training data and rebuild the model [34]. Prior to depicting the proposed method in this study, the basic notations and definitions of TL are briefly introduced as follows [34].

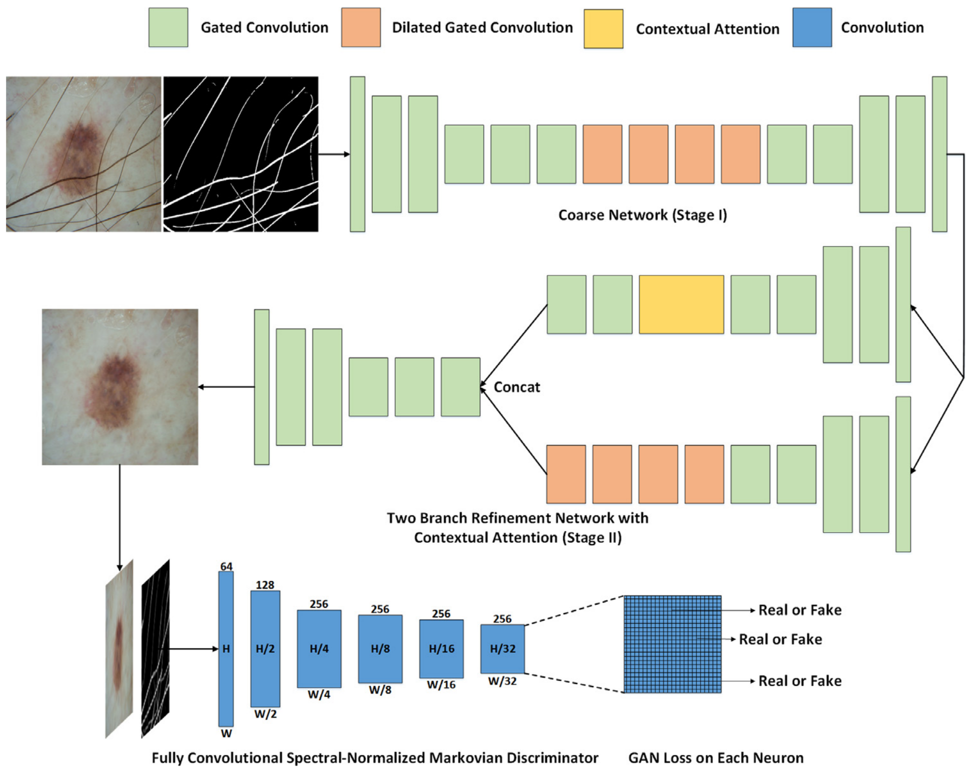

Fig. 6. Overview of the framework with gated convolution and SN-PatchGAN for free-form image inpainting.

Notation. A domain $\mathcal{D}$ consists of two components: a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$, in which $X = \{x_1, \ldots, x_n\} \in \mathcal{X}$. Given a specific domain $\mathcal{D} = \{\mathcal{X}, P(X)\}$, a task consists of two components: a label space $\mathcal{Y}$ and an objective predictive function $f(\cdot)$, denoted by $\mathcal{T} = \{\mathcal{Y}, f(\cdot)\}$, which is learned from the training data pairs $\{x_i, y_i\}$, where $x_i \in X$ and $y_i \in \mathcal{Y}$. The function $f(\cdot)$ is used to predict the corresponding label, $f(x)$, of a new instance x.

Definition. Given a source domain $\mathcal{D}_S$ and learning task $\mathcal{T}_S$, a target domain $\mathcal{D}_T$ and learning task $\mathcal{T}_T$, TL aims to help improve the learning of the target predictive function $f_T(\cdot)$ in $\mathcal{D}_T$ using the knowledge in $\mathcal{D}_S$ and $\mathcal{T}_S$, where $\mathcal{D}_S \neq \mathcal{D}_T$, or $\mathcal{T}_S \neq \mathcal{T}_T$.

The idea of TL is to use a network trained on a different domain for a different source task, and adapt it for other domain and target task. In this paper, due to the randomness of the hair shape, we use the pre-trained free-form image inpainting system proposed by Yu et al. [29] and trained on the CelebA-HQ (Celebrity face) dataset. The system combines Gated convolution and SN-PatchGAN as shown in Fig. 6 and the inpainting results demonstrate that this system is feasible for hair gap inpainting..

The Gated convolutions of the system learn from data through soft mask layers rather than hard-gated mask employed in traditional networks, as illustrated in Fig. 7. The process involves:

$$Gating_{y,x} = \sum \sum W_g \cdot I \tag{1}$$

$$Feature_{y,x} = \sum \sum W_f \cdot I \tag{2}$$

$$O_{y,x} = \phi(Feature_{y,x}) \odot \sigma(Gating_{y,x}) \tag{3}$$

where x, y represent the resolution of the output map, I and O are respectively the inputs and outputs, σ is sigmoid function which makes the output gating value between zero and one, ϕ can be any activation functions (e.g., ReLU, ELU and LeakyReLU), and finally W_g and W_f denote the two different convolutional filters [29]. In short, Gated convolution generalises partial convolution by providing a learnable dynamic feature selection mechanism for each channel at each spatial location across all layers. It also solves the issue of vanilla convolution which treats all input pixels as valid ones [29].

As free-form masks may appear anywhere in images with any shape, Yu et al. [29] present a patch-based GAN loss, called SN-PatchGAN, by applying spectral-normalised discriminator on dense image patches. SN-PatchGAN can inpaint multiple holes with any shape at any location and is different from other impainting networks which normally work by impainting single rectangular holes. Therefore, this method is suitable for inpainting a hair gap which has random shapes, and the results of hair gap inpainting are presented in Section 5.2. More details of this inpainting system are available in [29].

3.4. Intra-SSIM for DHR evaluation

From our experimentation we found that hair segmentation and hair gap inpainting must be iterated a few times to achieve complete removal of any hair strands. Interestingly, when we input the hair gap inpainted results of the first iteration into U-Net, the few hair strands that were not detected in the first iteration can be segmented out in the second iteration, as shown in Fig. 11. Likewise, repeating the above process enabled us to discover minute hair strands that improve the overall performance of the system. However, in order to determine the number of iterations which include hair segmentation and hair gap inpainting, we propose a method referred as Intra-SSIM to evaluate the results of DHR based on a single image without any reference to a hair free image.

The evaluation method we proposed is based on the structural similarity (SSIM) index. SSIM is introduced in detail in [35], which is summarized as follows. It is based on three comparison measurements between the samples of x and y, i.e., luminance (l), contrast (c) and structure (s). Let μ_x, σ_x^2, σ_{xy} be the mean of x, the variance of x, and the covariance of x and y, respectively. The com-

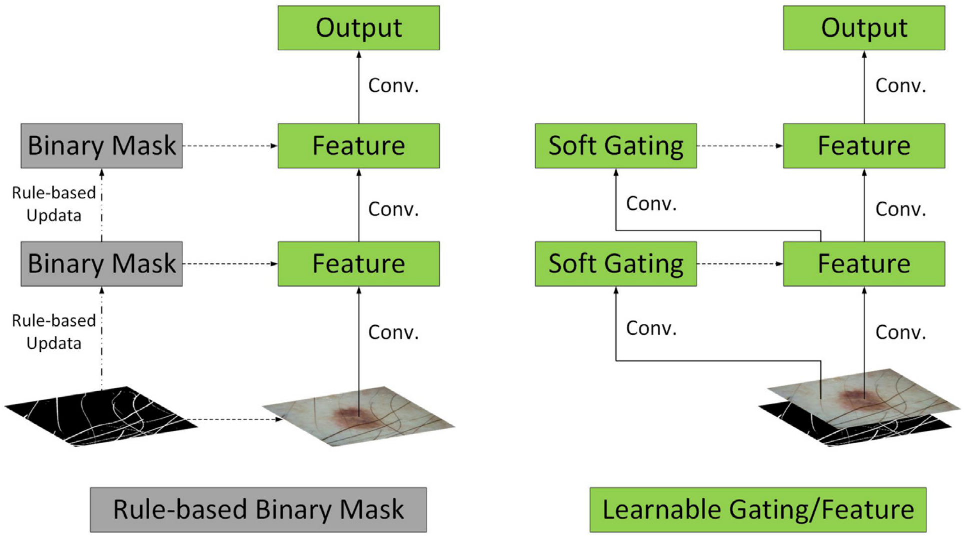

Fig. 7. Partial convolution (left): un-learnable single-channel feature hard-gating. Gated convolution (right): learnable dynamic feature selection mechanism for each channel at each spatial location across all layers.

parison measurements are:

$$l(x,y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}, \tag{4}$$

$$c(x,y) = \frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}, \tag{5}$$

$$s(x,y) = \frac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3}, \tag{6}$$

where c_1, c_2 and c_3 are small constants respectively given by

$$c_1 = (k_1 L)^2, \quad c_2 = (k_2 L)^2 \quad and \quad c_3 = c_2/2, \tag{7}$$

where L is the dynamic range of the pixel values ($L = 255$ for 8 bits/pixel grey scale images), and $k_1 = 0.01$ and $k_2 = 0.03$ are two scalar constants. The general form of the SSIM index between signal x and y of common size NxN is defined as

$$\mathrm{SSIM}(x,y) = \left[l(x,y)^{\alpha} \cdot c(x,y)^{\beta} \cdot s(x,y)^{\gamma}\right], \tag{8}$$

where α, β and γ are parameters to define the relative importance of the three components. Specifically, we set $\alpha = \beta = \gamma = 1$, and the resulting SSIM index is

$$\mathrm{SSIM}(x,y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \tag{9}$$

which satisfies the following conditions:

1. Symmetry: $\mathrm{SSIM}(x,y) = \mathrm{SSIM}(y,x)$;
2. Boundedness: $\mathrm{SSIM}(x,y) \le 1$;
3. Unique maximum: $\mathrm{SSIM}(x,y) = 1$ if and only if $x = y$.

The SSIM index is used for measuring the similarity between two images. It is a full reference metric, i.e., used for the measurement or prediction of image quality based on an initial uncompressed or distortion-free image as reference. However, in our case each hair image does not have the corresponding hair free image as a reference. Therefore, it is not feasible to use SSIM index to evaluate the image after DHR. To compute Intra-SSIM we first divide the image with the size of NxN pixels into nxn cell blocks. Let b $= \{b_m | m = 1, 2, \ldots, nxn - 1\}$ be the cell block, and let B $= \{B(i,j) | i, j = 1, 2, \ldots, n-3\}$ represent a 3x3 cell blocks where each cell block is (N/n)x(N/n) pixels. We then calculate the SSIM index between the central cell block and the remaining cell blocks in eight directions of each $B(i,j)$, i.e., SSIM_0, SSIM_1, SSIM_2, SSIM_3, SSIM_4, SSIM_5, SSIM_6, SSIM_7, and calculate the average value, i.e.,

$$\overline{B}(i,j) = \frac{1}{8}\sum_{m=0}^{7} \mathrm{SSIM\_}m. \tag{10}$$

Similarly, the sliding window technique with a stride of 1 along the row and column of the image is employed to calculate the Intra-SSIM of the whole image, i.e.,

$$\text{Intra-SSIM} = \frac{1}{(n-2)^2}\sum_{j=0}^{n-3}\sum_{i=0}^{n-3}\overline{B}(i,j). \tag{11}$$

In this paper, we divide the image of size 512x512 pixels into 32x32 cell blocks, so that each cell block comprises 16x16 pixels, as illustrated in Fig. 8. The process to compute Intra-SSIM is as follows:

Calculate Intra-SSIM of a inpainting image

Let $k = 32 \cdot j + i$
For $B(i,j)|_{i=29, j=29}$ **in** inpainting image:
$\overline{B}(i,j) = \frac{1}{8} SSIM[b_{k+33}, (b_k, b_{k+1}, b_{k+2}, b_{k+32}, b_{k+34}, b_{k+64}, b_{k+65}, b_{k+66})]$
Intra-SSIM $= \frac{1}{900}\sum_{j=0}^{29}\sum_{i=0}^{29}\overline{B}(i,j)$
return Intra-SSIM

The proposed method does not require a reference image and presents a new evaluation criterion based on a single image to examine the effects of the inpainting areas (e.g., hair gap) and the background (e.g., lesion). The experiment on Intra-SSIM is presented in Section 4.3.

4. Experiments

4.1. Implementation

We implemented the proposed DHR algorithm on Google Colaboratory with a computational capability of 3.7 Tesla K80 GPU. Google Colaboratory is a free online cloud-based Jupyter notebook environment which allows the training of deep learning models on CPUs, GPUs, and TPUs. The algorithms of Huang et al. [9] and Xie et al. [10] were implemented on MATLAB 2016b on a system with an Intel Core i5-4210U 1.7 GHz processor, while DullRazor [8] is a window software. There was no publicly available implementation for the other methods mentioned in this paper. We implemented the evaluation criteria on the same computer.

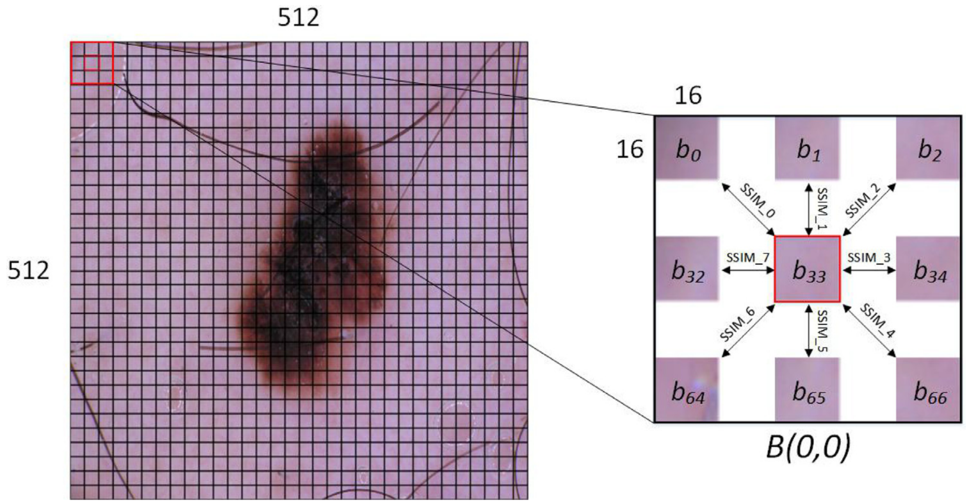

Fig. 8. Left shows the image with size of 512x512 pixels are divided into 32x32 cell blocks. At each step of the sliding window, the mean SSIM is computed for the central block by considering the eight neighbouring blocks as shown on right.

Table 2
The three steps of our experiment used the same testing data, where TL denotes transfer learning.

Procedure	Training	Validation	Testing	Image size
Hair segmentation	80	20	206	512x512
Hair gap inpainting	TL	TL	206	512x512
Lesion segmentation	1910	478	206	256x256

4.2. Datasets

Our data was extracted from the ISIC 2018: Skin Lesion Analysis Towards Melanoma Detection grand challenge datasets [36]. In the lesion boundary segmentation task of this challenge, the data consists of 2594 images and 2594 corresponding ground truth response masks. We manually selected 306 hair images with various hair types (i.e., various length, thickness and colour) and different types of artefacts, i.e., gel and tiny ruler markers). Subsequently, we created the 306 corresponding ground truth hair masks as explained in Section 3.1. In hair segmentation, we randomly selected 80 images for training, 20 images for validation and 206 images for testing. In hair gap inpainting, we used the same testing data. In skin lesion segmentation, we randomly selected 1910 images for training and 478 images for validation. We also used the same testing data. In hair segmentation and hair gap inpainting, we resized the images to 512x512 pixels. In skin lesion segmentation, for the ease of computation, we resized all the images to 256x256 pixels. The usage and distribution of data are shown in the Table 2.

4.3. Digital hair removal (DHR)

For better DHR, we repeated the two steps of hair segmentation and hair gap inpainting. To determine the number of repetitions, we calculated the average Intra-SSIM index of all testing images after each repetition. In order to verify whether the block size n affects the convergence, the proposed model was executed for $n =$ 16, 32, and 64. Our experiments show that varying n has no impact on the convergence, since for all n values the model presents a stable Intra-SSIM score after the fourth iteration. Interestingly, though the Intra-SSIM score has a slight increase with larger n, but since the average value of Intra-SSIM does not increase any further after the fourth iteration (see Fig. 9(b)), the inpainting process terminates and is therefore not affected by the block size. Also with an increase in n there is a significant increase in the computation time required for calculating Intra-SSIM, as shown in Fig. 9(a), and hence we reached a trade-off and chose n as 32 to achieve optimal performance.

To evaluate the inpainting procedure using the proposed Intra-SSIM technique, we selected 43 lighter and 43 darker images with hair over the lesions. The careful selection of images is required to demonstrate the effectiveness of the evaluation criterion for lesions with bright and dark textures. The experimental results show comparable results for both dark and lighter hair lesion images. For lighter hair images, after removal of hair the Intra-SSIM values improved from 0.487 to 0.682. Similarly for dark hair images, upon hair removal, the Intra-SSIM scores improved from 0.421 to 0.574. Fig. 10 shows an example of a lighter and dark hair image before and after hair removal using Intra-SSIM evaluation.

Furthermore, Fig. 11 illustrates the effectiveness of the model and presents the visualisation of the Intra-SSIM index. It is clearly seen that as the number of iteration increases the contours of the lesion becomes more visible, and the plane fitting indicates the volumetric shape of the lesion.

In our experiments, we used accuracy, Dice coefficient, specificity, F1 score, precision, sensitivity, Jaccard index (IoU) and Enhanced-alignment measure (E-meaure) to evaluate the performance of the segmentation techniques. Also, Intra-SSIM is used to evaluate the performance of inpainting while MSE is used to evaluate the degree of undesirable changes outside the hair gap within the lesions. In order to verify the importance of hair removal in the segmentation of skin lesions, we used five state-of-the-art methods to segment and compare the original image and the image after hair removal.

5. Results and comparison

The performance of our system was quantitatively and qualitatively evaluated in terms of hair segmentation, hair gap inpainting, and lesion segmentation. These results are compared with those generated by three state-of-the-art methods: DullRazor [8], Huang et al. [9] and Xie et al. [10].

5.1. Hair segmentation

5.1.1. Quantitative comparison of hair segmentation

The Jaccard index, also known as Intersection over Union, is a statistical measure for gauging the similarity and diversity of sample sets. The Jaccard coefficient measures the similarity between

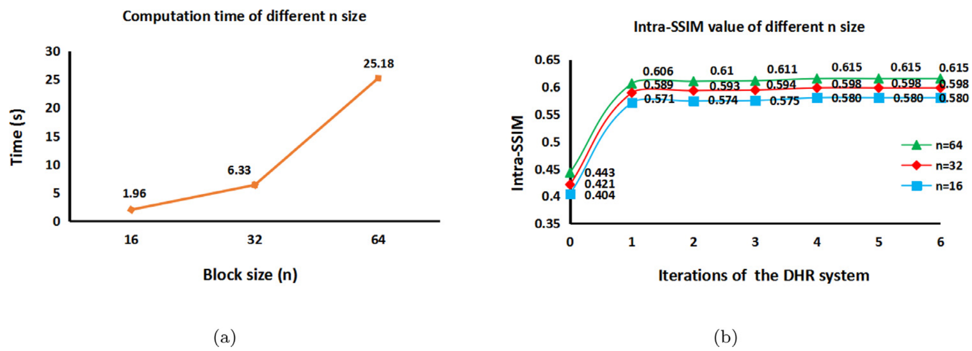

Fig. 9. (a) Computation time required for calculating Intra-SSIM for different block sizes n. (b) Average Intra-SSIM index computed for all test images with block sizes n of 16, 32 and 64.

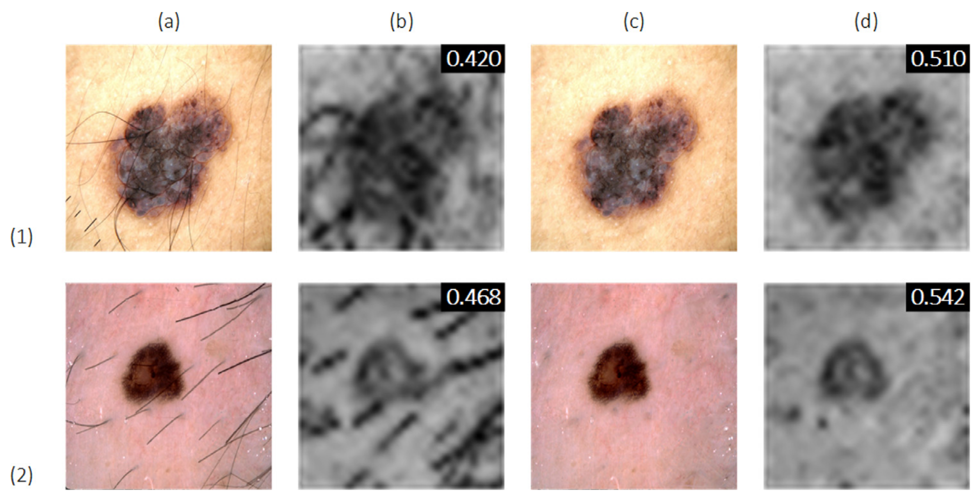

Fig. 10. Example of a lighter and dark hair images before and after hair removal using Intra-SSIM evaluation: Rows (1) and (2) respectively show the lighter and dark hair image. Along the column: (a) and (c) are the original image and the inpainting image, respectively; (b) and (d) are the Intra-SSIM visualizations corresponding to (a) and (c), respectively. The value of the Intra-SSIM of each image is also denoted.

two finite sample sets, and is defined as the size of the intersection divided by the size of the union of the sample sets, i.e.,

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}. \tag{12}$$

Simply put, the Jaccard index is the area of overlap between the predicted segmentation and the ground truth divided by the area of union between the segmentation result and the ground truth.

E-measure [37] is a recently proposed measure that consists of a compact term that simultaneously captures image level statistics and local pixel matching information. It is significantly better than traditional measures, e.g., F1 score. Also the metric presents an effective and efficient way for evaluating binary foreground maps. E-measure is computed as:

$$Q_{FM} = \frac{1}{w \times h} \sum_{x=1}^{w} \sum_{y=1}^{h} \phi_{FM}(x,y), \tag{13}$$

where ϕ_{FM} is enhanced alignment matrix, and h and w are the height and the width of the map, respectively. More details about E-measure are available in [37].

Table 3 shows that the performance of our hair segmentation method surpasses the other methods by a large margin. Our method achieves the best score in every evaluation metric, especially the Jaccard index. The proposed method records better Jaccard index for hair segmentation, 95.42 compared to 61.88 for DullRazor. One of the main advantages of the Jaccard index is the ability to penalise over- and under-segmentation [5]. The proposed method is able to improve the Jaccard index by discarding the false positives. Similarly, because of the accurate segmentation of hair, the proposed method achieves better E-measure for hair segmentation, namely 98.00 compared to 90.94 for DullRazor.

5.1.2. Qualitative comparison of hair segmentation

Fig. 12 shows the performance comparison for the proposed method with the three state-of-the-art methods. The figure shows DullRazor (refer to row 2) is able to remove black thick hair, but fails to detect the ruler marker and thick hair in images (2) and (3). The Huang et al. method (refer to row 3) produces a high rate of false positive in image (2) because it treats the hair-like texture as hair. Furthermore, it cannot fully detect the thick hair in image (3). The Xie et al. method (refer to row 4) fails to detect short hair in images (1) and (2), especially in image (1). For image (3), all three state-of-the-art methods segment the lesion texture as hair, which seriously affect the diagnosis of the lesion area by dermatologists. On the other hand, the proposed method (refer to row 5) accurately segments all kinds of hair and ruler markers.

5.2. Hair gap inpainting

5.2.1. Quantitative comparison of hair gap inpainting

The inpainting effects are evaluated using the proposed Intra-SSIM. After the hair strands are segmented, the segmented image is passed to the Gated convolution and SN-PatchGAN, where the segmented hair strands are morphologically filled. Next the average Intra-SSIM score of the inpainted image is computed and followed by the segmentation. The inpainting is repeated until the

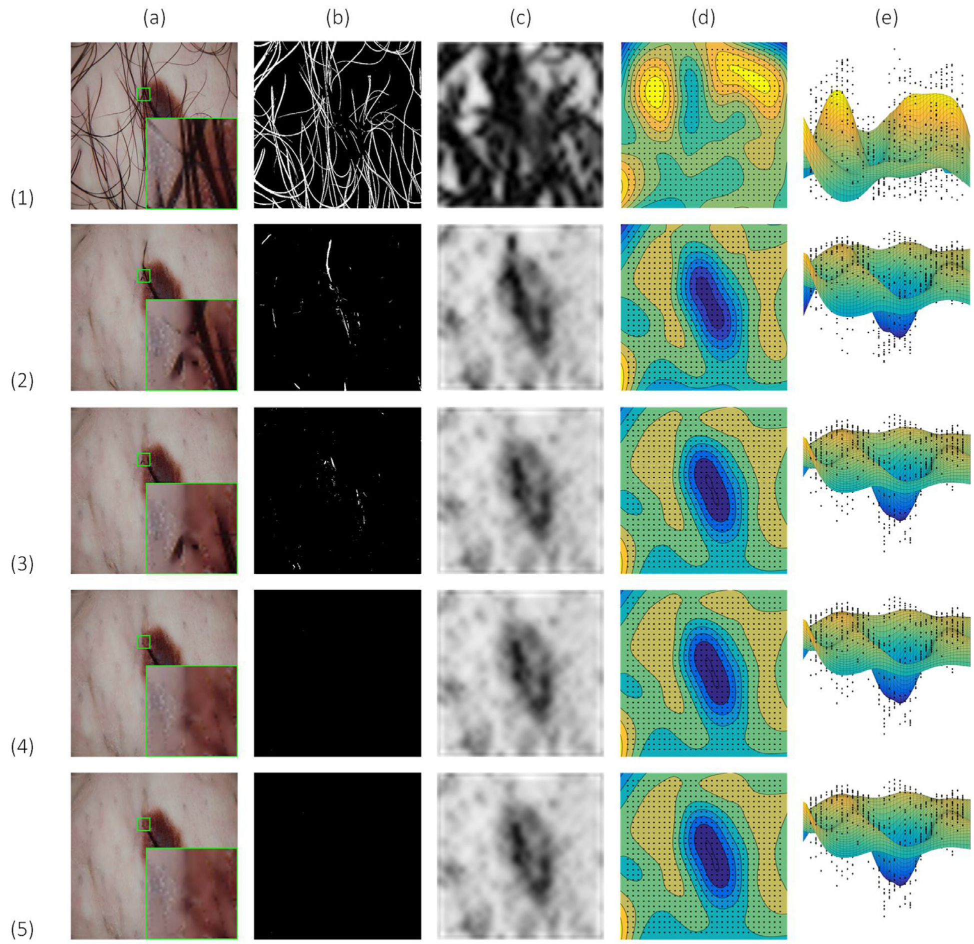

Fig. 11. Visualization of Intra-SSIM. Rows (1) (2) (3) (4) and (5) respectively show the input image, the image fed to the system after the first, second, third and fourth iterations. Along the column: (a) denote input images; (b) the segmented hair masks after each iteration; (c) visualization of the Intra-SSIM map after each iteration; (d) represents the contour map of the Intra-SSIM map; and (e) the plane fitted to the Intra-SSIM map.

Table 3
Comparison of hair segmentation results of four methods.

Segmentation method	Accu-racy	Dice-coef	Speci-ficity	F1-score	Prec-ision	Sensi-tivity	Jaccard-index	E-measure
DullRazor [8]	96.38	74.70	98.01	76.88	78.13	76.08	61.88	90.94
Huang et al. [9]	85.58	46.21	85.42	46.09	33.89	87.90	31.64	43.80
Xie et al. [10]	93.92	63.72	95.76	65.32	62.41	71.85	49.09	66.17
Proposed method	99.08	96.88	99.85	94.43	99.09	95.74	95.42	98.00

Intra-SSIM does not increase any further. In our experiments, as explained in Section 4.3, 4 iterations are required to achieve an acceptable hair gap inpainting.

To evaluate the performance of the hair gap inpainting, the Intra-SSIM score between the image inpainting areas and the background are computed for the different methods. For a realistic comparison we computed the inpainting results of each method on the hair test images using the hair masks we created as ground truths, and then used Intra-SSIM to evaluate the internal similarity between the hair gap inpainting results of each method and the background of the original image, as shown in Fig. 13. Furthermore, to measure any undesirable texture change within the lesion area due to poor segmentation and inpainting, the MSE of the lesion region of the input image and that of the inpainted image (excluding hair gap) is calculated, as shown in the Fig. 14. Since Xie et al. [10] did not propose a inpainting method, we compared the inpainting result with DullRazor [8] and Huang et al. [9].

Table 4 compares our method with two state-of-the-art methods. It can seen that the Intra-SSIM (the higher the better) and MSE value (the lower the better) for our method is the best. The inpainting results of our method are the most similar to the input image regardless of whether in the hair area or non-hair area.

Table 4
Comparison of inpainting results by three methods. Columns 3, 4 and 5 show the Intra-SSIM and the MSE for the images shown in Fig. 13 and Fig. 14. Column 6 presents an average scores for all the Test images.

Method	Metrics	Image (1)	Image (2)	Image (3)	Test Images
DullRazor [8]	Intra-SSIM	0.544	0.635	0.511	0.504
	MSE	40.7	39.4	13.8	4.91
Huang et al. [9]	Intra-SSIM	0.551	0.670	0.540	0.566
	MSE	92.6	53.4	63.3	10.27
Proposed method	Intra-SSIM	0.556	0.674	0.546	0.599
	MSE	2.6	3.6	1.8	0.65

Furthermore, to illustrate the effectiveness of the inpainting process (see Fig. 15), we added a hair mask (or a watermark mask) to a hairless image (or a watermark-free image) and then used dif-

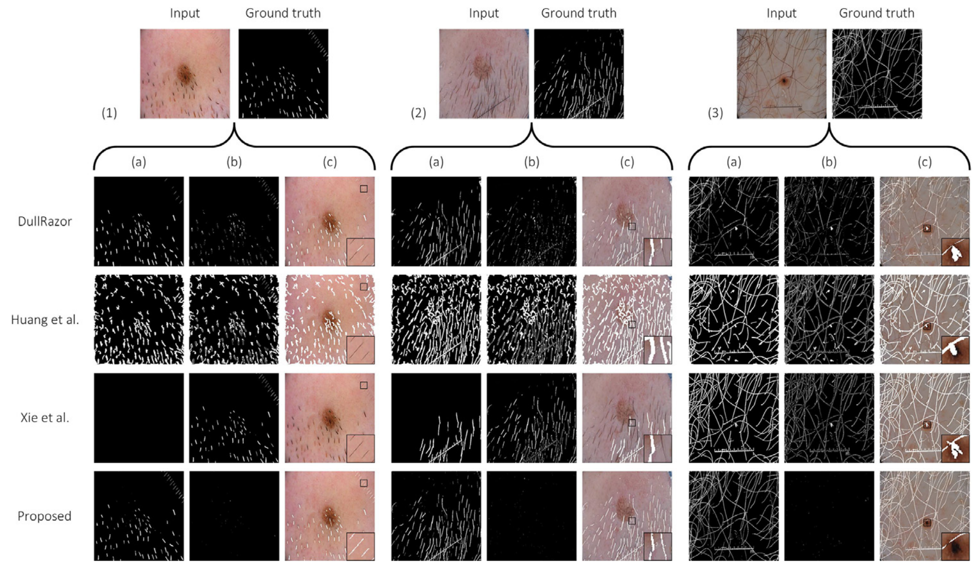

Fig. 12. Qualitative comparison of hair segmentation between the proposed method and with DullRazor [8]; Huang et al. method [9]; Xie et al. method [10]. Row 1 shows the input image and its corresponding ground truths, and rows 2,3,4 and 5 respectively show the segmentation results of the various methods. Column (a) Segmentation outputs of each method. Column (b) Differences between segmentation output and ground truth. Column (c) The segmentation result mask overlaid on the input image for detailed comparison.

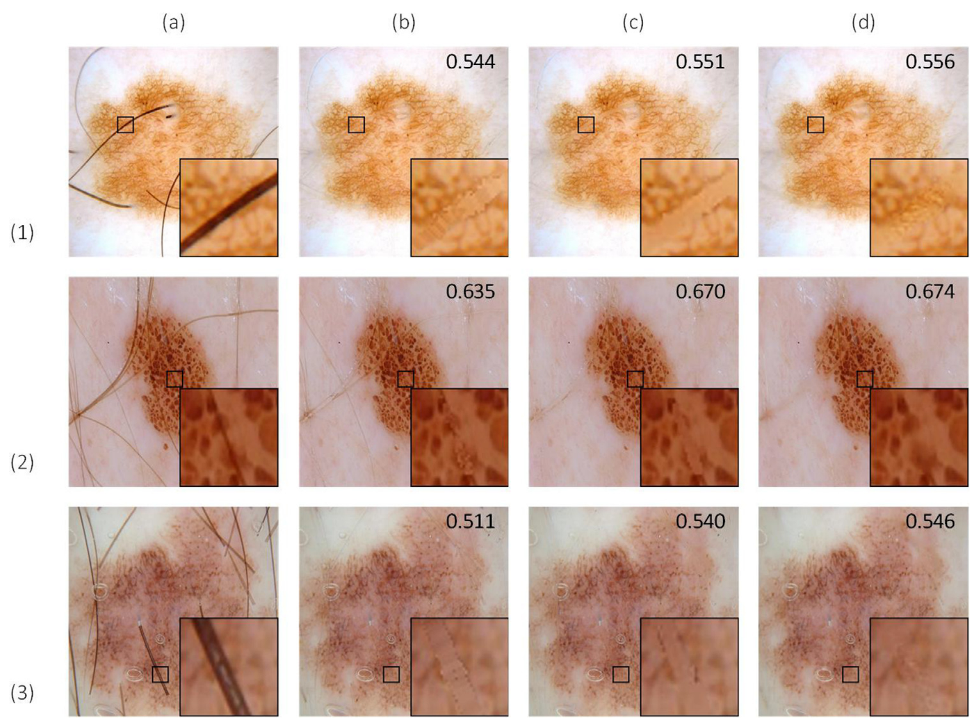

Fig. 13. Qualitative comparison of hair gap inpainting results: Column (a) Input images; Column (b) DullRazor [8]; Column (c) Huang et al. [9]; and Column (d) Proposed. The inset in each image shows the zoomed in enlargement of the hair gap inpainting results for clearer visualisation. The value of the Intra-SSIM of each image is also denoted.

ferent inpainting methods to inpaint it. Since DullRazor is an integrated software which does not allow customization, we compared our inpainting model with Huang's method. From our experimentation, the use of median filtering as an image inpainting method does not allow the inpainted area to blend well with the background features, leaving more obvious inpainting traces. Our method can inpaint the image according to the background features learned by the model, so that the inpainting results are closely integrated with the background.

5.2.2. Qualitative comparison of hair gap inpainting

We carefully selected three images from the dataset to illustrate the potential of the proposed method. These images contain hair strands which bisect the lesion area, and an effective DHR algorithm should remove the hair stands and accurately inpaint the hair gaps without any undesirable effects on the lesion area. Columns (b) and (c) of Fig. 13 respectively show the DHR results obtained using Bilinear interpolation and Median filtering. It can be easily verified visually that the inpainting area cannot match

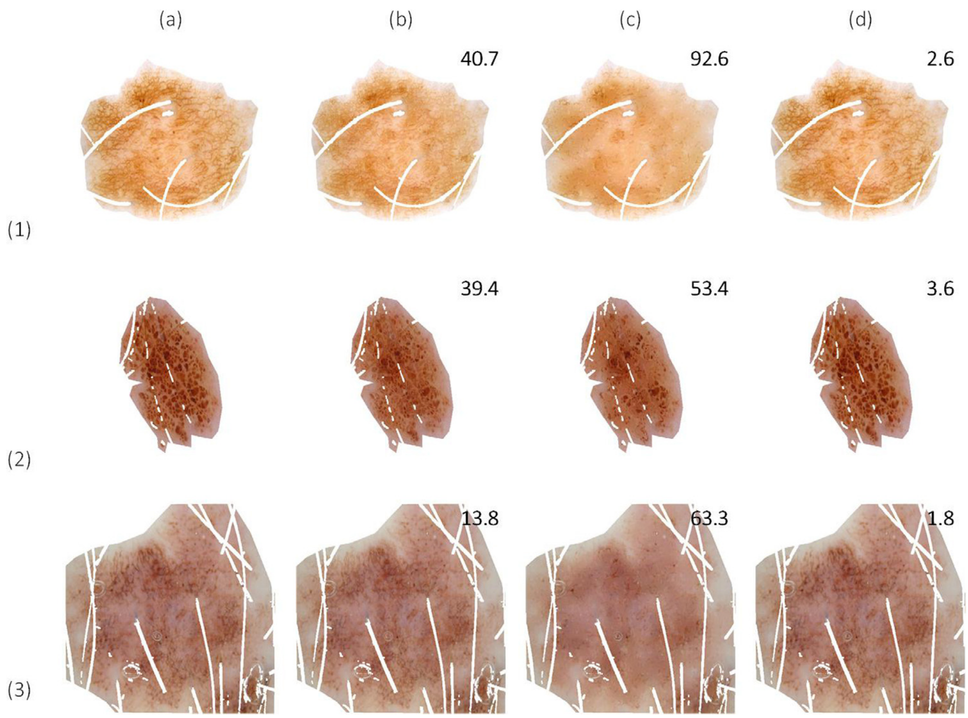

Fig. 14. Qualitative comparison illustrating the texture change within a lesion area. Column (a) Lesion of the input image (excluding hair gap). The results for inpainting using: Column (b) DullRazor [8]; Column (c) Huang et al. method [9]; and Column (d) Proposed. The value of MSE between the input and the inpainted lesion region is also denoted in each image.

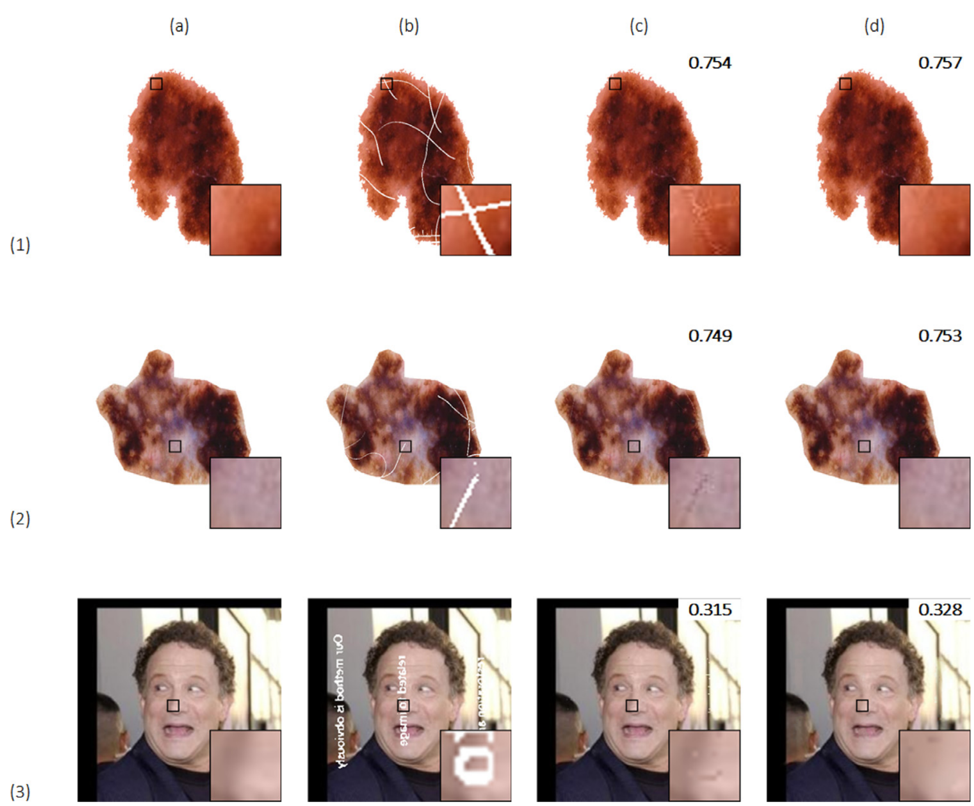

Fig. 15. Comparison of inpainting methods on hairless image (or watermark-free image): Column (a) original hairless images (or watermark-free image); (b) original image with added hair mask (or watermark); (c) and (d) are the inpainting results of Huang et al. [9] and proposed, respectively. The value of the Intra-SSIM of each image is also denoted.

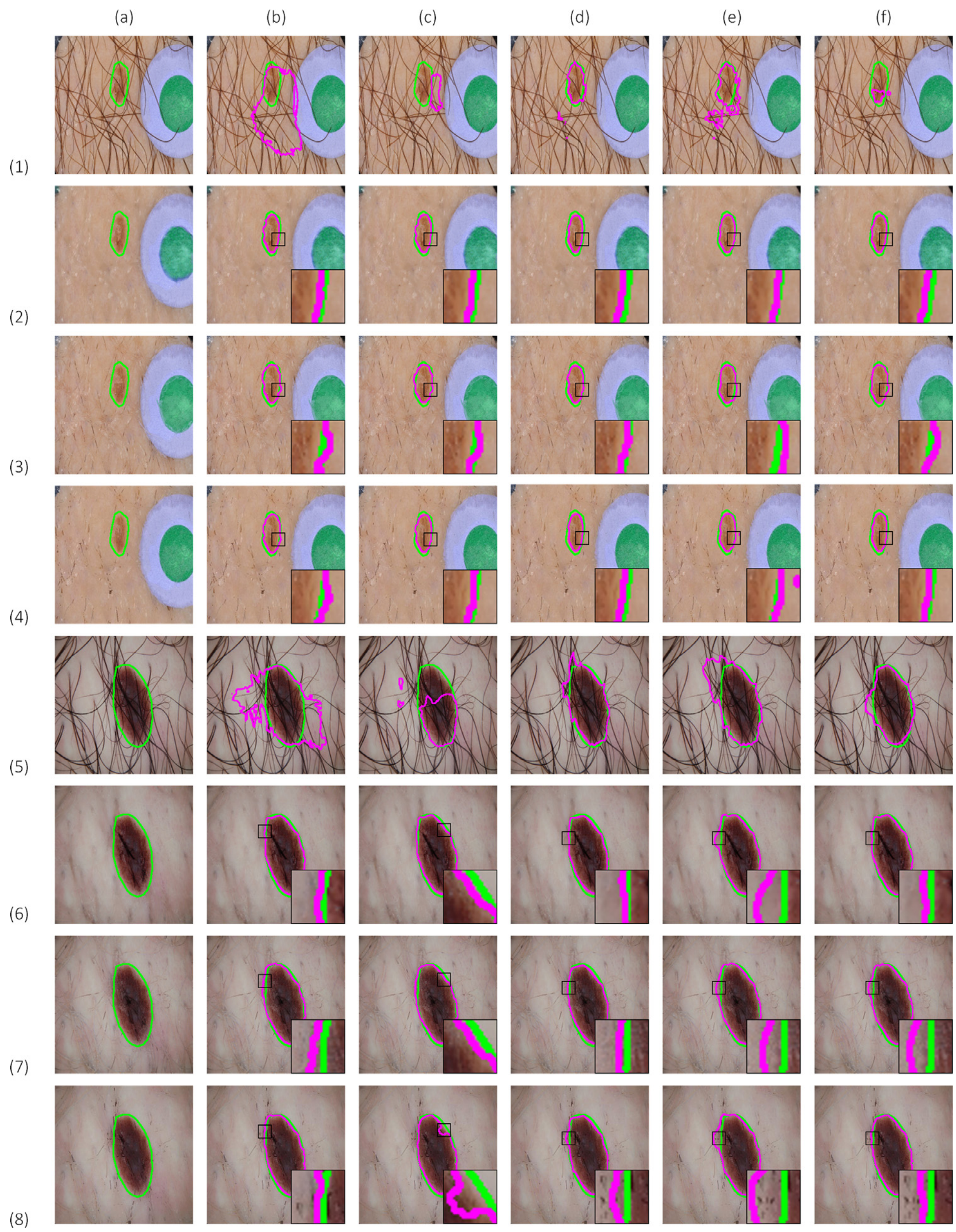

Fig. 16. Comparison of lesion segmentation before and after DHR. Rows (1) and (5) are the original images. Rows (2) and (6) are the corresponding inpainted results of the proposed method. Rows (3) and (7) are the corresponding inpainted results of DullRazor [8]. Rows (4) and (8) are the corresponding inpainted results of Huang et al. [9]. The green line denotes the ground truth, and the purple line denotes the results of the DHR methods. Column (a) are Input images. The results for lesion segmentation using: Column (b) AttU-Net [38]; Column (c) UNet++ [39]; Column (d) R2AttU-Net; Column (e) R2U-Net [40]; and Column (f) U-Net [28]. Our proposed method makes the lesion area clear, improving the accuracy of diagnosis and analysis of lesions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the input image texture. Furthermore, due to poor segmentation and inpainting effects, the texture characteristics of the lesion area are considerably changed, causing the lesion area to be blurred, as shown in Fig. 14.

Our method which comprises of segmentation through U-Net, inpainting using Gated convolution and SN-PatchGAN, and evaluation using Intra-SSIM performs better than Bilinear interpolation and Median filtering employed respectively by DullRazor [8] and Huang et al. [9]. It almost retains the original texture features of the lesion area. Thus, our inpainting effect is more natural without obvious inpainting track and more similar to the original image. The inpainting results are closer to the original texture and hardly lose the information on the lesion.

5.3. Comparison of skin lesion segmentation with original and hair removed images

5.3.1. Quantitative comparison of lesion segmentation

We used the same training dataset to train the following five state-of-the-art segmentation methods: AttU-Net [38], UNet++ [39], R2AttU-Net, R2U-Net [40] and U-Net [28]. We used the input image and the corresponding inpainted image to test them. Table 5 shows the test results. After application of hair removal as a pre-processing step, the average value of each evaluation metric has been improved, which shows that hair removal is very important for the segmentation of skin lesions. In addition, we compared the results of lesion segmentation after different hair removal meth-

Table 5
The result of skin lesion segmentation on the input image and the inpainted image for the proposed method.

Method	Input	Accu-racy	Dice-coef	Speci-ficity	F1-score	Prec-ision	Sensi-tivity	Jaccard-index	E-measure
AttU-Net [38]	Original	91.98	85.26	95.38	85.07	87.42	85.31	76.39	88.75
	Inpainted	93.05	87.49	96.09	88.61	90.18	88.20	79.74	90.81
UNet+ [39]	Original	91.45	83.03	97.96	82.34	91.26	77.17	74.02	86.72
	Inpainted	92.35	86.13	97.91	85.73	93.11	80.85	77.62	88.98
R2AttU-Net	Original	91.38	84.07	98.02	83.27	91.91	77.76	74.93	87.17
	Inpainted	91.72	84.79	98.58	83.94	93.95	77.30	75.91	87.54
R2U-Net [40]	Original	92.03	85.34	95.71	84.88	87.0	85.38	76.60	89.10
	Inpainted	92.78	86.72	96.94	86.40	90.10	85.10	78.64	89.98
U-Net [28]	Original	92.35	85.77	97.38	85.70	91.52	81.86	77.17	89.00
	Inpainted	93.07	87.42	96.80	87.77	91.19	86.65	79.43	90.58
Average improvement		**+0.76**	**+1.82**	**+0.37**	**+2.24**	**+2.02**	**+2.12**	**+2.45**	**+1.43**

Table 6
Comparison of the average segmentation results of different hair removal for the segmentation methods AttU-Net [38], UNet++ [39], R2AttU-Net, R2U-Net [40] and U-Net [28].

DHR Method	Accu-racy	Dice-coef	Speci-ficity	F1-score	Prec-ision	Sensi-tivity	Jaccard-index	E-measure
DullRazor [8]	92.56	86.36	97.38	86.38	91.42	83.61	78.01	89.56
Huang et al. [9]	92.47	86.36	97.29	86.36	91.36	84.09	78.05	89.46
Proposed method	92.59	86.51	97.26	86.49	91.71	83.62	78.27	89.58

ods. Table 6 shows the average values of each evaluation metric of different hair removal methods on the five segmentation methods. Our proposed method performed the best on the main segmentation evaluation metrics (Dice coefficient, Jaccard index and E-measure).

5.3.2. Qualitative comparison of lesion segmentation

Without changing the texture characteristics of the lesion area, the edge of the lesion area can be more obvious by application of hair removal, which makes it more conducive to the recognition of the lesion area by the method. Fig. 16 shows the lesion segmentation before and after DHR. When using hair images, the messy hairs make it difficult for the segmentation method to distinguish the edge of the lesion area, resulting in poor segmentation. After removing the hair, the lesion area in the image becomes clear, and the segmentation method easily distinguishes the lesion area from the skin background. When compared with other hair removal methods, we find that our method is more accurate and the boundary between the lesion and the background is more obvious. Thus, our segmentation result is better. More importantly, good hair removal is more helpful for dermatologists to diagnose skin diseases.

6. Conclusion

In this paper, we propose a hair removal method based on deep learning. In order to train the U-Net model and obtain accurate hair segmentation results, we created a hair mask dataset and made it publicly available for other researchers to use and jointly promote the development of the field. For the hair gap inpainting work, we employed transfer learning techniques, and proposed a new single-image non-reference inpainting evaluation method referred as the Intra-SSIM. The method presents a new evaluation index for non-reference inpainting work by calculating the similarity within the image to evaluate the inpainting results. The output of the proposed method is more accurate and of higher quality than three other state-of-the-art methods. We also verified the importance of hair removal for lesion segmentation. In addition, our proposed framework can also be used to precisely remove specific objects, e.g., mesh and watermarks on images. We believe that our hair removal method can aid in the diagnosis of skin cancer. Our contributions include:

- We are the first to use deep learning to complete the challenging task of DHR in dermoscopic images;
- Creating a hair mask dataset to solve the problem of shortage of hair mask dataset;
- We also propose a non-reference SSIM-based single image DHR evaluation method, Intra-SSIM.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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